On the relevancy of the Turbulent Dispersed Flows as a Phenomenological and Conceptual Model of the Financial Markets Microstructure
A Kinetic approach of the Financial Markets.

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Abstract

Here we discuss the relevancy of the turbulent dispersed flows (TDF) as a conceptual model of the financial markets (FM) microstructure.
An historical overview brings out the ontological realism and the analogy and deduction from the phenomena as the common philosophical modes of both phenomena. Then we discuss the foundations of an analogy: (i) an empirical basis of this analogy is established by giving the empirical similarities between both phenomena (ii) simple interactions between the basic components of the FM ([3], p.30) are commented using main characteristics of the TDF ([4], sect.2): dispersed topology, Stokes number, particle loading, equilibrium estate, (iii) a compounded investigation of agent-agent correlations is run from a molecular chaos assumption for the closure of colliding particles velocities distribution in TDF([12], p.6).
Subsequently, a kinetic approach is proposed for the modeling of the FM dynamics through the building of a kinetic equation on a probability distribution function.

Keyword: Market Microstructure, Turbulent Dispersed Flows, Analogy, Kinetic approach

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